(ii) Dwarf shoot or branches of limited growth

These branches do not have apical buds and hence show only limited growth. They develop in the axils of scale leaves and bear both scale and foliage leaves.

Leaves

There are two types of leaves 1. scale leaves, 2. foliage leaves

1. Scale leaves:

They are dark, brown, membranous, thin and small. They are present on both long and dwarf shoots. Their function is to protect young buds. The scale leaves on the dwarf shoots have a distinct midrib and are called "**Cataphylls**".

2. Foliage leaves:

The foliage leaves are green angular and needle like structures. They are borne on the dwarf shoot. A dwarf shoot with a group of needle like foliage leaves is known as **foliar spur**. The number of needles per dwarf shoot varies among the species. It may be one (*Pinus monophylla*), two (*P. sylvestris*), three (*P. geraradiana*), four (*P. quadrifolia*) and five (*P. excelsa*).

Internal Structure

T.S. of root

The internal structure of root reveals the presence of epiblema, cortex and stele.

The epiblema is made up of single layer of parenchymatous cells. Cortex is the wide zone and consists of parenchyma. Some of the cells have resin ducts. A single layered endodermis with suberised wall is present and is impregnated with tannins.A multilayered pericycle made up of parenchyma is present. Vascular tissue is radial, diarch with exarch xylem. The protoxylem bifurcates to form a 'Y' shaped structure and a resin duct lies in between the two arms of protoxylem. Secondary growth is present (Figure 2.49).

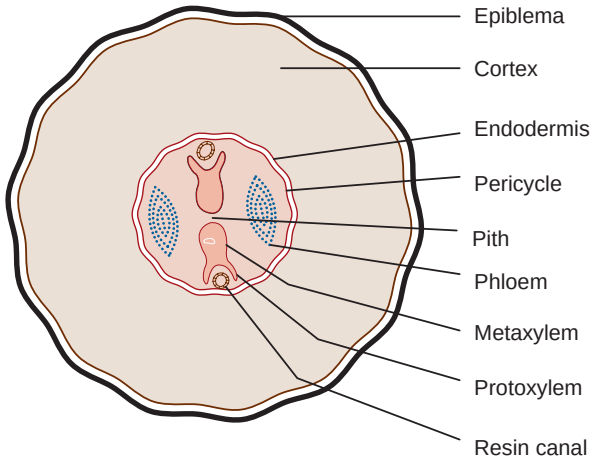


Figure 2.49: T.S. of *Pinus* root

T.S. of Stem

The internal organization of the stem shows three regions namely epidermis, Cortex and vascular tissue (Figure 2.50).

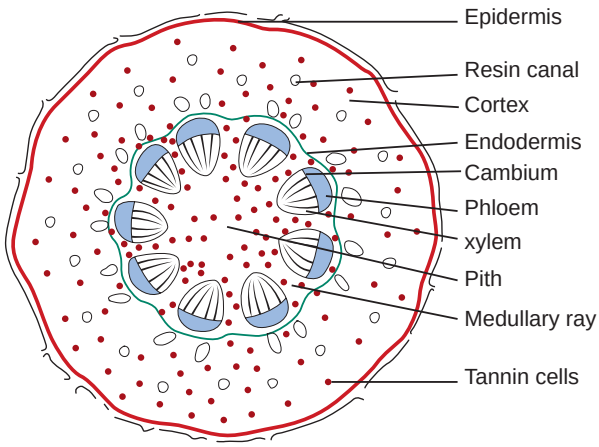


Figure 2.50: T.S. of *Pinus* stem

Epidermis is the outermost layer composed of compactly arranged and heavily cutinized cells. Epidermis is followed by few layers of sclerenchymatous hypodermis. The cortex consists of thin walled parenchyma cells. Resin canals and tannin filled cells are present in this